# LALITH ALPURI

**AI INFRASTRUCTURE ENGINEER · MCP SERVERS · LOCAL AGENT MEMORY**

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## PROFILE

AI infrastructure engineer and applied AI strategist building local-first systems for coding agents and decision support. I work across Python, Java, MCP servers, SQLite memory, hybrid retrieval, multi-agent simulation, market intelligence, predictive analysis, and full-stack delivery. My public work includes EpisodAI, Episoda Core MCP, ALPURIS OS, and a personal Sentinel / UWOF local AI lab.

## TECHNICAL STRENGTHS

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| **AI systems** | MCP servers, AI agents, multi-agent orchestration, local LLM workflows, RAG, hybrid retrieval, knowledge graphs, context engineering |
| **Languages** | Python, Java, JavaScript/ES6+, SQL, Bash, R |
| **Backend** | Spring Boot, Spring, JPA/Hibernate, REST APIs, Node.js, SQLite/FTS5, MySQL, MongoDB |
| **Frontend** | React, Next.js, HTML, CSS, WebGL visualization |
| **Delivery** | Docker, Kubernetes, Git/GitHub, GitHub Actions, Postman, Selenium automation, manual QA, HLD/LLD, DSA |
| **Local AI** | MLX, Ollama, OpenInterpreter, LoRA fine-tuning, Apple Silicon model tiers, Obsidian workflows |

## PROFESSIONAL EXPERIENCE

### Ataraxia Pharma Pvt. Ltd. | Applied AI, Simulation & Analytics | 2026

- Developed AI-assisted research and scenario-analysis workflows for wellness product planning, combining structured market inputs, synthetic consumer cohorts, and comparative evidence reviews.
- Worked with a 1,000,000-row Monte Carlo simulation dataset in SQLite; analyzed pricing, channel commissions, landed costs, net selling price, and margin variables through feasibility and profitability filters.
- Produced a ranked 1,000-scenario CSV export and an HTML simulation explorer to support inspection of model assumptions, trade-offs, and shortlisted scenarios.
- Applied predictive scenario analysis to acquisition cost, ROAS, conversion, channel mix, and unit economics; distinguished synthetic outputs and forecast assumptions from observed commercial performance.
- Structured competitor comparisons and scientific-evidence audits into spreadsheets, dashboards, and technical briefs, documenting source evidence, limitations, and validation requirements.

## PROJECTS

### EpisodAI | Creator / maintainer | 2025–Present

- Built a local MCP memory engine with SQLite WAL, vector and trigram retrieval, knowledge-graph workflows, cognitive decay, and Obsidian ingestion.
- Hardened the threaded HTTP bridge for concurrent requests and documented reproduction commands instead of relying on unverified benchmark claims.

### Episoda Core MCP | Creator / maintainer | 2025–Present

- Built a zero-dependency Python MCP server using SQLite FTS5/BM25 for offline, persistent recall by coding agents.
- Designed a small proof path—save, search, stats, and tests—to keep the system auditable and easy to adopt.

### ALPURIS OS | Creator / maintainer | 2025–Present

- Built a local multi-agent city simulation with autonomous citizens, sector activity, civic governance, an economy, Python services, and a 3D WebGL voxel interface.
- Added repo-relative configuration, portable workspace/database overrides, Docker/Compose support, and a reproducible demo path.

### Sentinel / UWOF local AI systems lab | Personal research project | 2025–Present

- Designed a local orchestration runner that routes between MLX model tiers, a web-scout path, and an OpenInterpreter/Ollama execution path.
- Maintains local model checkpoints and an MLX LoRA fine-tuning script for Apple Silicon experiments; runtime assumptions are documented in the source folder.

## FULL-STACK ENGINEERING

### Independent Full-Stack Developer | 2025–2026

- Built web applications with React/JavaScript frontends and Spring Boot backends, following SOLID principles and clear API boundaries.
- Designed SQL schemas and Hibernate/JPA mappings; built REST endpoints with global exception handling and Postman validation.
- Applied manual QA and Selenium automation to improve regression coverage and release confidence.

## EDUCATION & CERTIFICATION

### Java Full Stack Development Certification | KodNest, Bengaluru

Java, Python, Spring Boot, JPA/Hibernate, React, SQL, MongoDB, REST APIs, DSA, manual QA, and Selenium automation.

### Bachelor of Commerce in Computer Application | Rayalaseema University, Andhra Pradesh

Coursework included Python, Core Java, C/C++, R, DBMS, and enterprise application foundations.

## SELECTED LINKS

GitHub: https://github.com/lalithbuilds · Project home: https://lalithbuilds.github.io/ · LinkedIn: https://www.linkedin.com/in/lalith-alpuri/